

Physics - Undergraduate Level Assessment 25

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is

- A. $mc/2$
- B. $mc/(2^{1/2})$
- C. mc
- D. $(3^{1/2})mc$

Question 2

(2 marks)

De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?

- A. Planck's constant
- B. Boltzmann's constant
- C. The Rydberg constant
- D. The speed of light

Question 3

(2 marks)

A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is

- A. independent of M
- B. proportional to $m^{1/2}$

C. linear in R

D. proportional to $R^{3/2}$

Question 4

(2 marks)

A refracting telescope consists of two converging lenses separated by 100 cm. The eye- piece lens has a focal length of 20 cm. The angular magnification of the telescope is

A. 4

B. 5

C. 6

D. 20

Question 5

(2 marks)

A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?

A. 0.25 W

B. 0.5 W

C. 1 W

D. 4 W

True / False

Question 1

(2 marks)

In a reversible thermodynamic process, the temperature of the system must remain constant throughout the process.

True

False

Question 2*(2 marks)*

Electromagnetic radiation emitted from a nucleus is often in the form of microwaves.

Question 3*(2 marks)*

The speed of light in a nonmagnetic dielectric material with a dielectric constant of 4.0 is 3.0×10^8 m/s.

Question 4*(2 marks)*

The Hall coefficient can be used to determine the sign of charge carriers in a doped semiconductor semiconductor.

Question 5*(2 marks)*

The negative muon, μ^- , has properties most similar to those of the electron, including charge and lepton classification.

Long Form**Question 1***(10 marks)*

Discuss the similarities between the emission spectrum of doubly ionized lithium (Li^{++}) and hydrogen, focusing on the wavelength changes and the underlying physics principles.

Type your answer here...

Question 2

(10 marks)

A tube of water is moving at $1/2 c$ relative to a lab frame. Analyze how the speed of a light beam, entering the tube in the same direction, is affected by the index of refraction of water.

Type your answer here...

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